

BarangayLink v2

Start

Prerequisites

Make sure you have these installed: - Node.js (v18 or higher) - npm or pnpm - A code editor (VS Code recommended)

Step 1: Install Dependencies

```
npm install
```

Step 2: Set Up Your Accounts

You'll need accounts for these services:

Service	Purpose	Sign Up Link
Convex	Database & Backend	convex.dev
Clerk	User Authentication	clerk.com
Mapbox	Maps (Optional)	mapbox.com
Resend	Email Sending (Optional)	resend.com
Liveblocks	Real-time Features (Optional)	liveblocks.io

Step 3: Create Your Environment File

Copy the template and fill in your keys:

```
cp .env.template .env.local
```

Open `.env.local` and add your API keys. At minimum, you need:

```
# Required
NEXT_PUBLIC_CONVEX_URL=https://your-project.convex.cloud
NEXT_PUBLIC_CLERK_PUBLISHABLE_KEY=pk_test_XXXXX
CLERK_SECRET_KEY=sk_test_XXXXX
```

Step 4: Set Up Convex Database

```
npx convex login
```

```
npx convex dev
```

Environment Variables Explained

Required Variables

Variable	Where to Get It	Description
NEXT_PUBLIC_CONVEX_URL	Convex Dashboard → Settings	Your Convex deployment URL
NEXT_PUBLIC_CLERK_PUBLIC_KEY	Clerk Dashboard → API Keys	Public key for authentication
CLERK_SECRET_KEY	Clerk Dashboard → API Keys	Secret key (keep private!)

Optional Variables

Variable	Purpose
NEXT_PUBLIC_MAPBOX_ACCESS_TOKEN	Enable map features
RESEND_API_KEY	Send emails (invitations, notifications)
LIVEBLOCKS_SECRET_KEY	Real-time document collaboration
NEXT_PUBLIC_LIVEBLOCKS_PUBLIC_KEY	Real-time features (client-side)

Project Structure

```
barangaylink-v2/
  src/
    app/                                # Next.js pages and routes
      dashboard/                         # Main dashboard
      projects/                         # Project management
      events/                           # Event management
      admin/                             # Admin pages
      portal/                           # Resident portal
      components/                       # Reusable UI components
      lib/                              # Utility functions
    convex/                             # Backend (database, API)
      schema.ts                         # Database schema
      users.ts                          # User operations
      projects.ts                       # Project operations
      events.ts                         # Event operations
    public/                             # Static files
  package.json
```

User Roles

The system has 5 user roles with different permissions:

Role	Access Level
ADMIN	Full system access, manage all users and settings
CAPTAIN	Full access except system settings
MANAGER	Manage projects and events in their department
BUILDER	Create and edit tasks, assigned projects
WORKER	View and complete assigned tasks only

Features Overview

Dashboard

- Overview of projects, events, and tasks
- Quick stats and recent activity
- Team workload visualization

Projects

- Create projects with budgets and timelines
- Assign team members
- Track milestones and progress
- Generate reports (PDF/Excel)

Events

- Community event scheduling
- Task assignment and tracking
- QR code attendance system
- Guest registration

Admin Panel

- User management and approvals
- Department configuration
- System settings and backups
- Certificate management

Resident Portal

- Public event viewing
- Certificate requests
- Community announcements

Tech Stack

- **Frontend:** Next.js 15, React 19, TailwindCSS
 - **Backend:** Convex (serverless database + functions)
 - **Auth:** Clerk
 - **UI:** Radix UI, Lucide Icons, Framer Motion
 - **Maps:** Mapbox GL
 - **PDF:** jsPDF, React-PDF
 - **Charts:** Recharts
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